



Machine Learning (ML)

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1. Introduction

One of the most exciting things about computers right now is machine learning (ML). Over the past few decades, ML has become a routine part of everyday life and has been able to tackle real-world problems. ML may be applied in a lot of different areas, like business, engineering, medical, and more. There are a lot of different kinds of learning algorithms in ML. Some of the most common include decision trees, linear regression, k-nearest neighbors, support vector machines, and neural networks. Some newer methods are also available, such as boosted tree models and deep learning Logistic Regression (tuned), best for transparency and accountability, high interpretability, lower risk of hidden bias, and easier to audit. Random Forest: Good accuracy, more complex, harder to explain than logistic regression, but still explainable with feature importance and partial dependence. Stronger performance can reduce harm from errors, but you must add explainability and bias checks. XGBoost is often the highest performance, but typically the least intuitive, and more sensitive to tuning and leakage risks. High performance can reduce false negatives or false positives, but you must compensate with transparency tools, calibration, and strong monitoring. In practice, it's hard to find the correct settings and architecture for ML models so that the learner model works effectively for both learning and generalizing. When it comes to using ML in real life, there are even more problems, such as dealing with a lot of missing, distorted, or unclear data. In addition, people need to be able to understand ML methods if they are going to be deployed in real life. Interpretability helps individuals comprehend how an ML model works and makes them more sure of the outcomes it gives. The first issue of "Applied Machine Learning" had 41 research papers in it. The papers focus on how to use ML models in different fields and problems. They discuss how to think about problems, how to describe data, how to build features, how to build ML models, and how to compare them critically with other methods. They also discuss how to understand the results.

2. Literature Review

The foundational literature and frameworks of machine learning delineate a rapidly evolving landscape of paradigms and technologies:

- **Core Paradigms:** The best way to group the area is by how algorithms learn. Supervised learning teaches models how to do classification and regression by using datasets with labels. Unsupervised learning finds hidden patterns in data that don't have labels. For example, it can be used to group people into groups based on their interests. **Reinforcement learning (RL)** is when agents learn how to make the best choices by getting rewards and punishments while they are in an environment that is always changing.
- **Important Research and Milestones:** The 2017 book "Attention Is All You Need" was a big step forward in the field. It brought in the **Transformer architecture**, which changed how NLP and computer vision work. Other big breakthroughs are **ResNet** for recognizing images and the rise of **foundation models**, which are huge networks trained on huge datasets that are used as a base for certain applications.
- **Frameworks and Technologies:** The **JAX AI stack** has become a fast numerical core for training the best models, while **TensorFlow** and **PyTorch** are still the most popular tools for research and production, respectively.
- **Educational Context:** Books like **Oliver Theobald's** "Machine Learning for Absolute Beginners" and **Aurélien Géron's** "Hands-On Machine Learning" give practitioners a solid theoretical and hands-on introduction to the field.

3. Analysis

Machine learning improves on traditional statistical and rule-based methods in both qualitative and quantitative ways:

- **Accuracy and Scale:** Deep learning can look at entire raw records, such as electronic health records (EHR), to accurately predict medical events like in-hospital death

(AUROC 0.93–0.94) (Rajkomar et al., 2018). This is not like traditional models that need to be manually feature-engineered.

- **Efficiency:** The new AI-based climate model **DL-ESy-M** can show how the climate has changed over the past 1,000 years in just **12 hours** on one processor. According to the University of Washington (2025), a regular supercomputer would take 90 days to complete the same thing.
- **Structural Reasoning: Neuro-symbolic AI** improves traditional "black box" neural networks by combining perception with **symbolic reasoning**. This allows systems explain and back up their choices, which is highly crucial in industries like law and finance that are tightly regulated (Colelough & Regli, 2025).
- **Multimodal Fusion:** Systems that use more than one sort of data, such as audio and visual cues, can cut down on false positives by **60–80%** compared to systems that only use one type of input. This is because one kind of data can help make sense of another (Openstf.io, 2025).
- **Confidence-Driven Data Augmentation:** Frameworks like **ADLC** make models more accurate by discovering predictions that aren't very certain and adding more, better data points to the dataset to make it easier to generalize (Sunaryono et al., 2025).

4. Applications

Machine learning is changing many different industries around the world right now through specific use cases (Nitin Rane on October 24, 2024):

- **Healthcare:** ML helps find diseases early, make personalized treatment plans, and **label radiology images automatically** to deal with large backlogs of images.
- **Finance:** Algorithms can predict **churn** and help banks reduce customer loss by spotting early warning signals in customer activity.
- **Manufacturing:** The field makes **predictive maintenance** possible by predicting when machines will break down before they do, which can cut downtime by up to 50%.
- **Agriculture:** ML makes **precision farming** possible by analyzing soil data, keeping an eye on crop health, and predicting yields automatically
- **Transportation:** Systems help **self-driving cars** find their way, save fuel by finding the best routes, and manage traffic intelligently.
- **Environmental Action:** AI-powered early warning systems keep communities safe from bad weather, and maps that **show high carbon stock** help with reforestation (United Nations Framework Convention on Climate Change [UNFCCC], 2025).

5. Ethical Considerations

ML has some good points, but it also raises important moral issues:

- Fairness and bias

One of the most important moral issues in AI and ML is fairness and bias. Machine learning algorithms learn from past data, and if that data has biases, the algorithms can keep those biases going and even make them worse. For instance, facial recognition systems have been shown to make more mistakes with people with darker skin tones. This can lead to unfair treatment, discrimination, and violations of privacy.

To fix bias in AI, we need datasets that are diverse and representative, thorough testing, and algorithms that are made to reduce bias. It is also important to keep an eye on and make changes to the system over time to make sure that machine learning ethics are followed.

- Keeping your information safe and private

There are worries about privacy and data security because AI models need a lot of data to train. There are a lot of responsibilities that come with collecting and storing personal data. Individuals and businesses can face serious problems if they get into systems without permission, leak data, or use sensitive information inappropriately. To deal with these ethical problems in machine learning, AI developers and companies need to put data protection first, use strong encryption, and follow privacy laws like the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). Individuals must be able to control their own personal information, so data-use policies must be clear and strong consent mechanisms must be in place.

- Being open and clear

AI and ML models are often "black boxes," which means it's hard to know how they make decisions. When AI systems are used in important areas like healthcare or finance, where understanding why a decision was made is very important, this lack of transparency can be a problem.

To alleviate this issue, researchers are creating AI models that are easier to understand and ways to explain AI decisions. Transparency and explainability not only help users trust the system, but they also make it easier to find and fix any mistakes or biases that might be there. These efforts help make machine learning ethics more responsible by doing so.

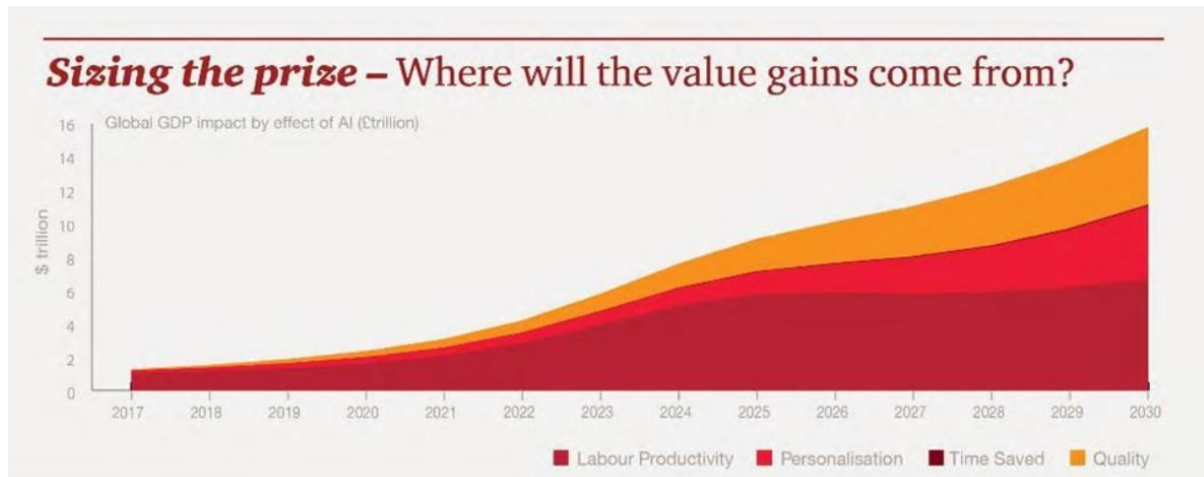
- Being responsible and accountable

As AI and ML systems become more independent, people start to wonder who is responsible and accountable. If an AI system makes a bad choice, who is to blame? Is it the AI, the company that is putting the system in place, or the developer?

To deal with these ethical issues in machine learning, it is important to set clear lines of responsibility and accountability. We need to make laws and rules that spell out who is responsible and make sure that developers and companies do what they need to do to keep AI systems from hurting people.

Forty-eight percent of businesses use machine learning (ML), data analysis, and AI tools to keep their data accurate.

AI is expected to contribute \$15.7 trillion to the global economy by 2030, more than the current output of China and India combined(*Forbes Advisor India, 2024*).



- Job Displacement and Economic Impact

The widespread adoption of AI and automation technologies has the potential to displace jobs in various industries. While AI can create new opportunities and increase productivity, it can also lead to job loss and economic disruption for specific groups. Artificial intelligence (AI) could replace the equivalent of 300 million full-time jobs, a report by investment bank [Goldman Sachs](#) says: “It could replace a quarter of work tasks in the US and Europe but may also mean new jobs and a productivity boom”.

Ethical considerations here involve addressing the societal impact of automation by investing in retraining and upskilling programs for affected workers, developing policies that promote job transition, and ensuring that the benefits of AI are distributed equitably.

AI and ML technologies offer immense potential for innovation and progress in various fields. However, to harness their benefits while minimizing harm, it is essential to address the ethical considerations surrounding their development and deployment. Fairness, transparency, accountability, privacy, and economic impact are among the moral dimensions that require careful consideration.

Responsibly developing and using AI and ML requires collaboration among technologists, policymakers, ethicists, and society. By prioritizing ethical considerations, we can ensure that

AI and ML systems enhance human well-being and contribute positively to our future. As these technologies continue to evolve, an ongoing commitment to ethics will remain crucial in guiding their development and application.

6. Practical Implementation

❖ Dataset name: Bank Customer Churn Prediction.

- Info.

Dataset shape (rows, columns): (10000, 14)

Dataset size, samples (rows): 10000

Dataset size, total columns: 14

Number of features, raw predictors (exclude target): 13

Number of features, after dropping ID columns: 10

Number of features, after one hot encoding: 13

Number of classes (Exited): 2

Class counts:

Exited

0 7963

1 2037

Name: count, dtype: int64

❖ Data Expo.

Columns:

['RowNumber', 'CustomerId', 'Surname', 'CreditScore', 'Geography', 'Gender', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'HasCrCard', 'IsActiveMember', 'EstimatedSalary', 'Exited']

Missing values per column:

RowNumber 0

CustomerId 0

Surname 0

CreditScore 0

Geography 0

Gender 0

Age 0

Tenure 0

Balance 0

NumOfProducts 0

HasCrCard 0

IsActiveMember 0

EstimatedSalary 0

Exited 0

dtype: int64

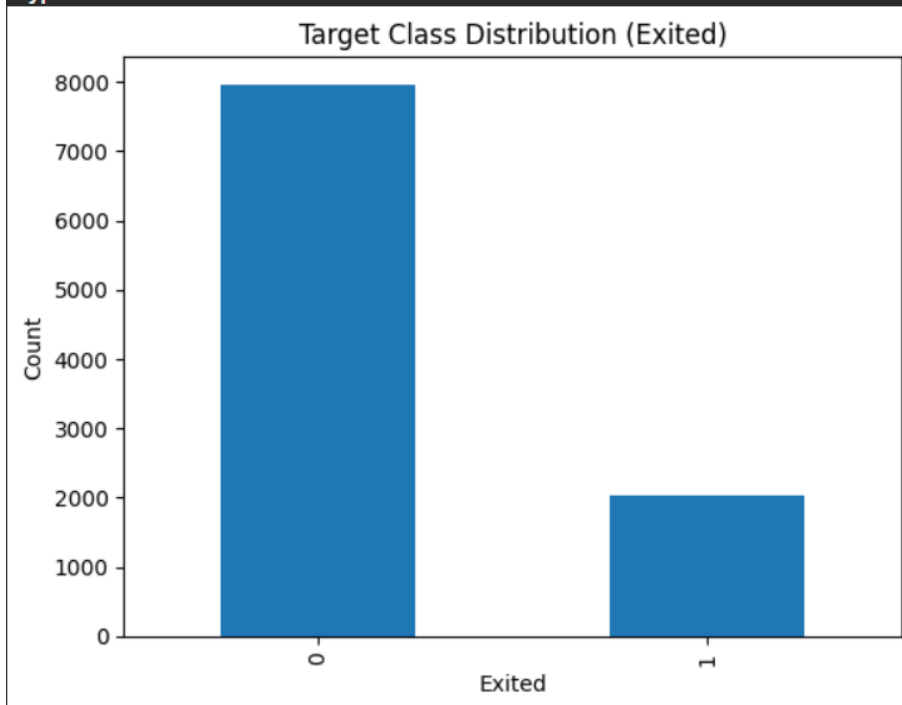
Describe numeric columns:

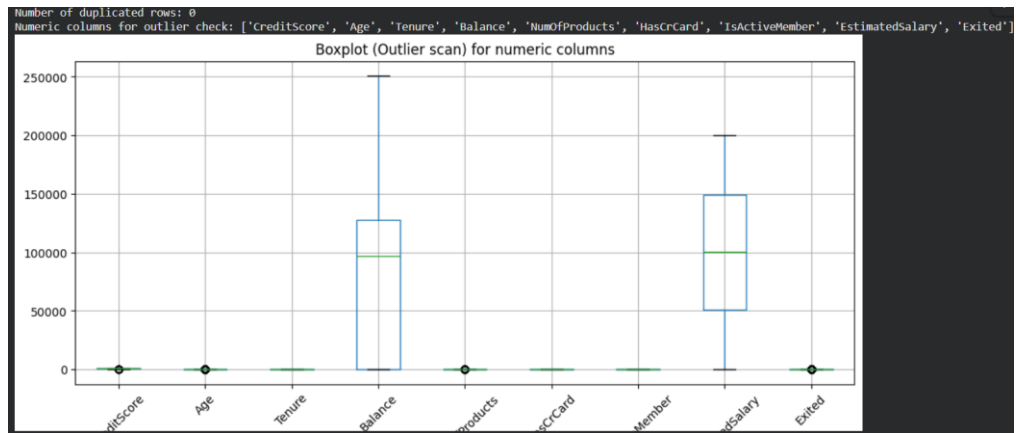
Index ▼	count	mean	std	min	25%	50%	75%	max
Tenure	10000.0	5.0128	2.892174377049708	0.0	3.0	5.0	7.0	10.0
RowNumber	10000.0	5000.5	2886.8956799071675	1.0	2500.75	5000.5	7500.25	10000.0
NumOfProducts	10000.0	1.5302	0.5816543579989936	1.0	1.0	1.0	2.0	4.0
IsActiveMember	10000.0	0.5151	0.4997969284589202	0.0	0.0	1.0	1.0	1.0
HasCrCard	10000.0	0.7055	0.45584046447514237	0.0	0.0	1.0	1.0	1.0
Exited	10000.0	0.2037	0.40276858399484955	0.0	0.0	0.0	0.0	1.0
EstimatedSalary	10000.0	100090.239881	57510.49281769922	11.58	51002.11	100193.915	149388.2475	199992.48
CustomerId	10000.0	15690940.5694	71936.18612274883	15565701.0	15628528.25	15690738.0	15753233.75	15815690.0
CreditScore	10000.0	650.5288	96.65329873613061	350.0	584.0	652.0	718.0	850.0
Balance	10000.0	76485.889288	62397.40520238623	0.0	0.0	97198.54000000001	127644.24	250898.09
Age	10000.0	38.9218	10.487806451704591	18.0	32.0	37.0	44.0	92.0

Target distribution:

	count
Exited	
0	7963
1	2037

dtype: int64





Outlier counts per numeric column:

	0
Exited	2037
Age	359
NumOfProducts	60
CreditScore	15
Tenure	0
Balance	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0

dtype: int64

❖ Preprocessing

Final feature matrix shape: (10000, 13)

❖ Split the data (Train, Validation, Test)

Train: (6000, 13) Val: (2000, 13) Test: (2000, 13)

❖ Model 1, Logistic Regression

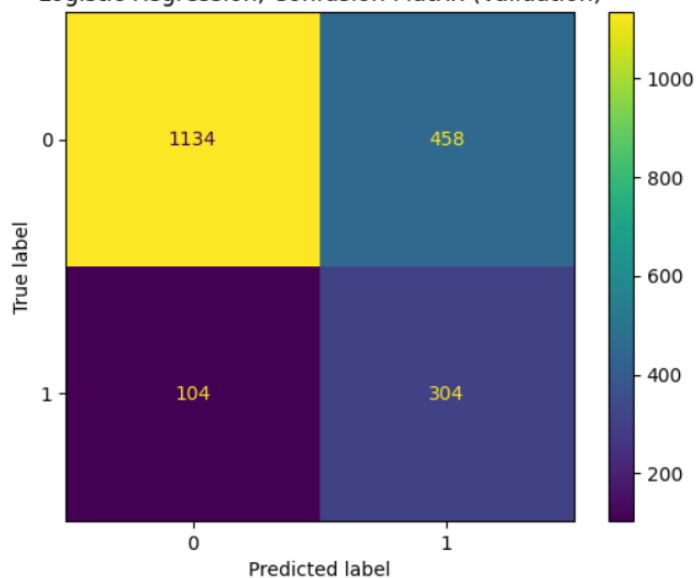
Validation metrics (Logistic Regression)

Accuracy: 0.719
 Precision: 0.399
 Recall: 0.7451
 F1: 0.5197
 ROC AUC: 0.7939
 Log Loss: 0.5769

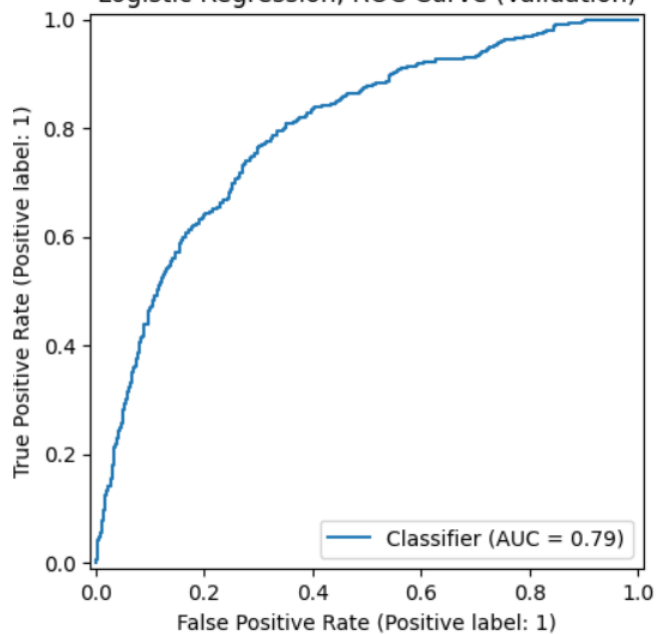
Classification report:

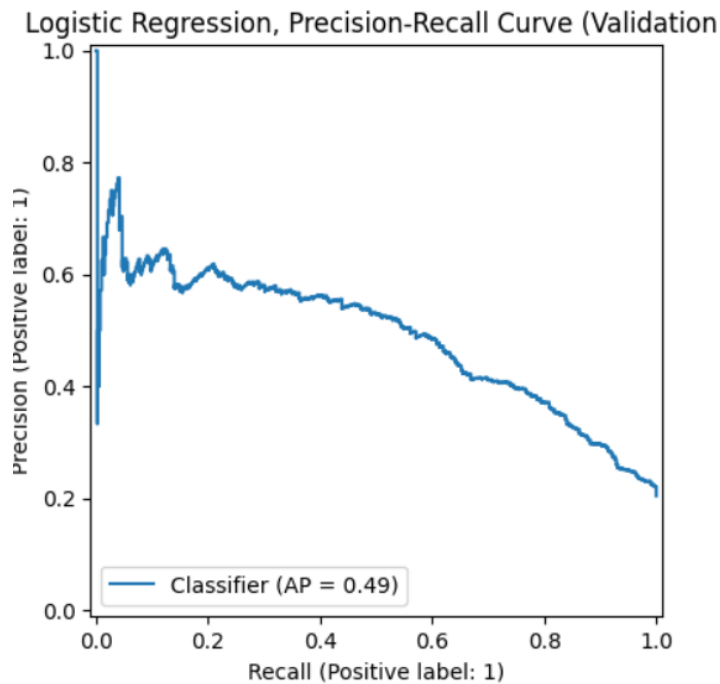
	precision	recall	f1-score	support
0	0.9160	0.7123	0.8014	1592
1	0.3990	0.7451	0.5197	408
accuracy			0.7190	2000
macro avg	0.6575	0.7287	0.6605	2000
weighted avg	0.8105	0.7190	0.7439	2000

Logistic Regression, Confusion Matrix (Validation)



Logistic Regression, ROC Curve (Validation)





❖ Model 2, Random Forest

Validation metrics (Random Forest)

Accuracy: 0.8595

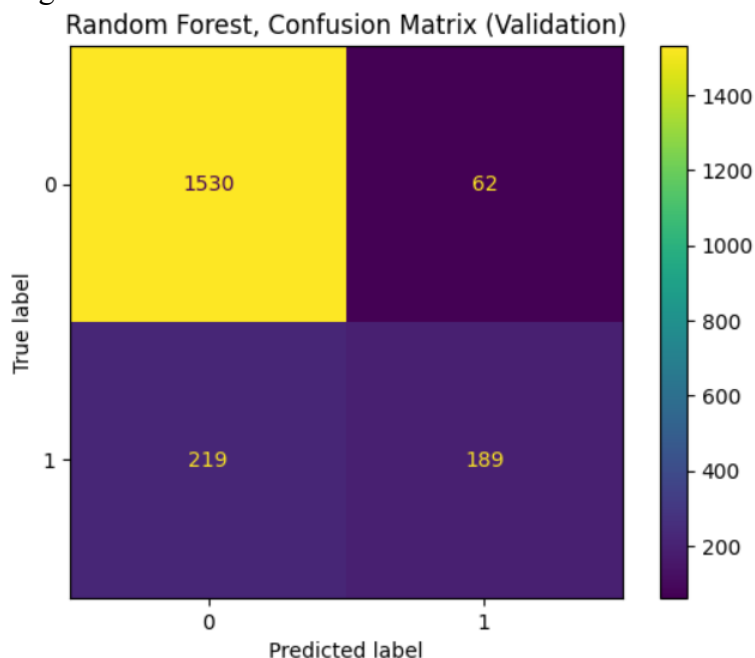
Precision: 0.753

Recall: 0.4632

F1: 0.5736

ROC AUC: 0.8635

Log Loss: 0.3434



❖ Model 3, XGBoost and tuning

Validation metrics (XGBoost)

F1: 0.6127 ROC AUC: 0.865

Fitting 5 folds for each of 15 candidates, totalling 75 fits

Best tuned parameters: {'subsample': 1.0, 'reg_lambda': 1.0, 'n_estimators': 400, 'max_depth': 5, 'learning_rate': 0.03, 'colsample_bytree': 0.9}

Tuned XGBoost F1 (Validation): 0.6241

Best model on validation: Tuned XGBoost , F1 = 0.6241

❖ Final evaluation on Test set

TEST metrics (Best sklearn model: Tuned XGBoost)

Accuracy: 0.81

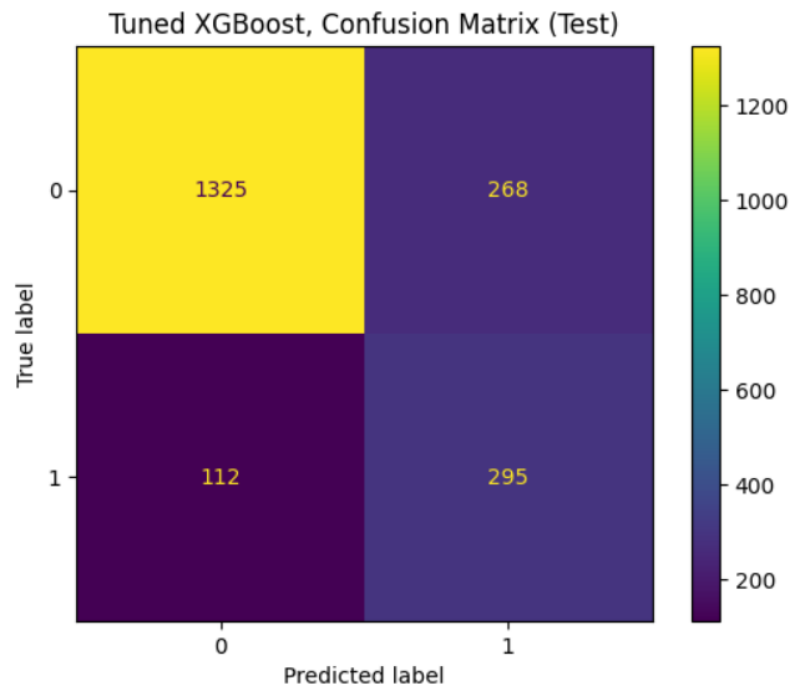
Precision: 0.524

Recall: 0.7248

F1: 0.6082

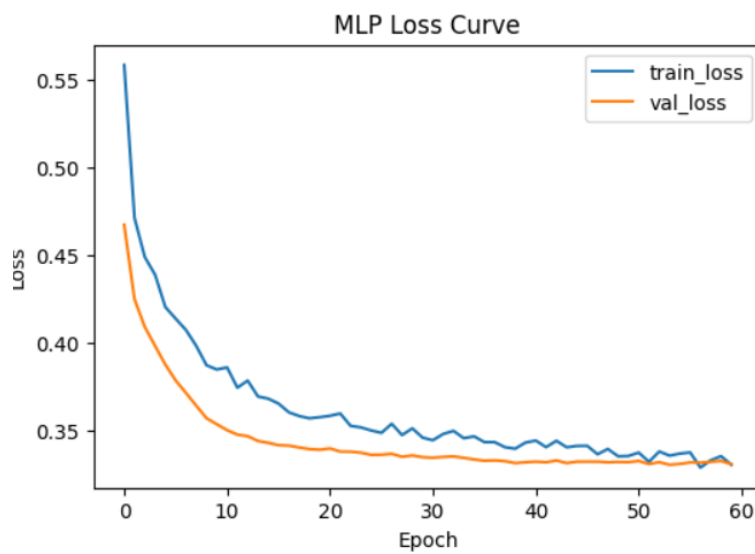
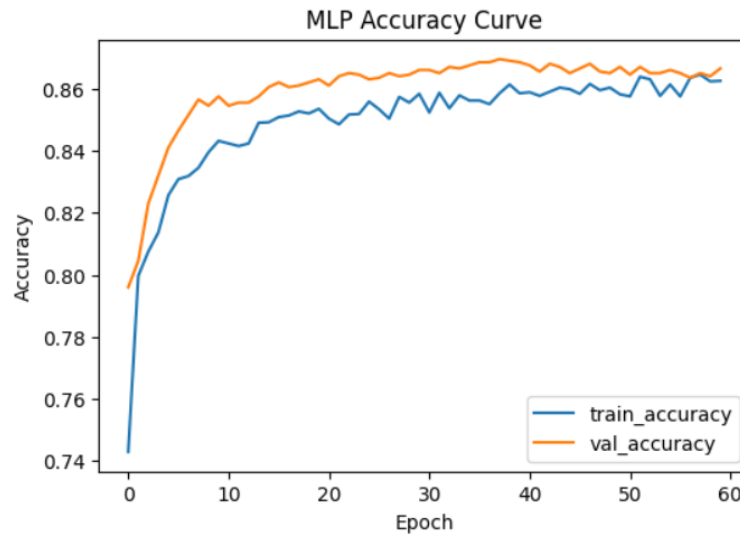
ROC AUC: 0.8554

Log Loss: 0.4263



❖ accuracy and loss curves

Model: "sequential"		
Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	896
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 32)	2,080
dropout_1 (Dropout)	(None, 32)	0
dense_2 (Dense)	(None, 1)	33



TEST metrics (MLP)

Accuracy: 0.857

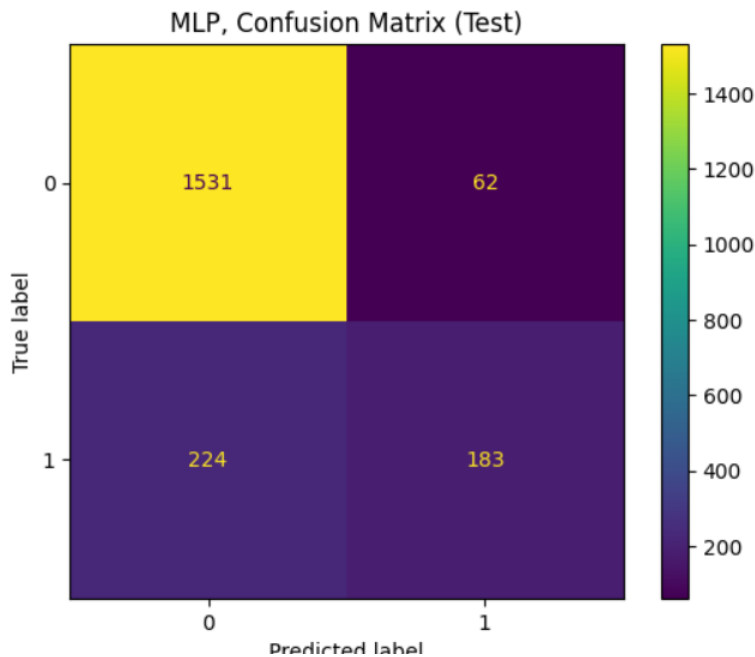
Precision: 0.7469

Recall: 0.4496

F1: 0.5613

ROC AUC: 0.8553

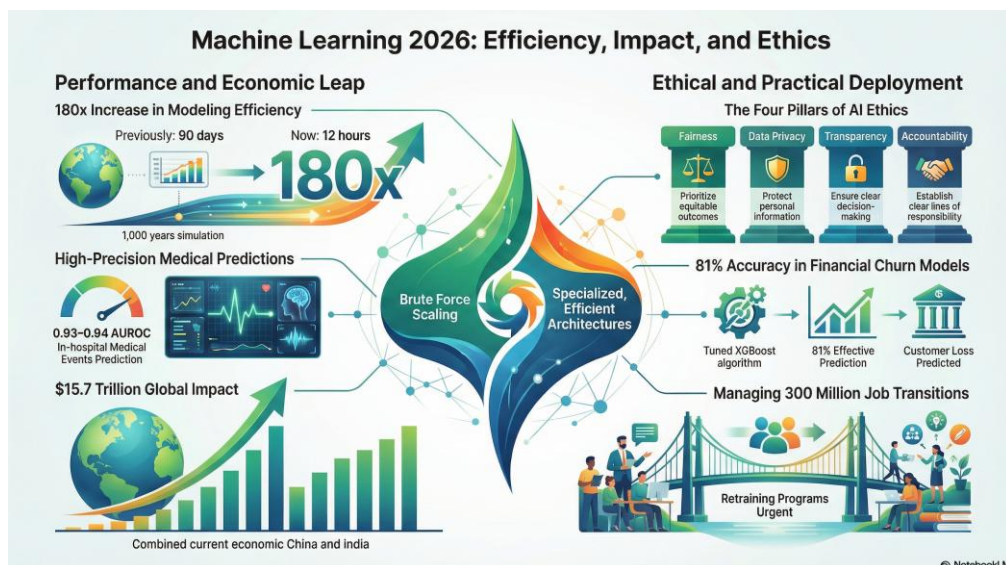
Log Loss: 0.3427



7. Conclusion

The state of machine learning in 2026 is characterized by a transition from "brute force" scaling to **efficient, specialized architectures**. Key findings indicate that future success relies on **Explainable AI (XAI)** to foster stakeholder trust and **Edge AI** to enhance intelligence accessibility on local devices.

Future directions encompass the amalgamation of **Quantum-AI hybrids** for expedited material discovery and the emergence of **Small Language Models (SLMs)** that provide elevated accuracy with a diminished environmental impact. As AI agents become digital coworkers, companies need to set up **agentic governance** to make sure these self-driving systems stay safe, open, and in line with human values.



→A video to clarify the research for those who love visuals ([click me to watch](#))

References

Initially, the research used AI tools (as helpers), such as Grammarly and QuillBot, to check grammar. It helped us check for errors, correct misspellings, improve punctuation, detect plagiarism, and enhance clarity. Additionally, we used Google [NotebookLM](#) for research and looked for references, also to generate a conclusion image and a video that illustrates the idea, which we supplemented with our own ideas and research.

[Applied Machine Learning](#)

Raschka, S., & Mirjalili, V. (n.d.). *Machine learning paradigms*. Wolfram Research.

<https://www.wolfram.com/language/introduction-machine-learning/machine-learning-paradigms/>

Building production AI on Cloud TPUs with JAX

Rajkomar, A., Oren, E., et al. (2018). *Scalable and accurate deep learning with electronic health records*. *npj Digital Medicine*.

<https://doi.org/10.1038/s41746-018-0029-1>

University of Washington. (2025, August 25). *This AI model simulates 1000 years of the current climate in just one day* [Press release]. UW News.

<https://www.washington.edu/news/2025/08/25/ai-simulates-1000-years-of-climate/>

Colelough, B. C., & Regli, W. (2025). *Neuro-Symbolic AI in 2024: A systematic review*.

CEUR Workshop Proceedings.

<https://ceur-ws.org/Vol-3819/paper3.pdf>

Openstf.io. (2025, November 10). *Why multimodal AI is the next big leap*.

<https://openstf.io/?p=33>

Sunaryono, D., Sarno, R., Sabilla, S. I., Putri, R. A., Amri, T. C., Mirda, I., & Siswantoro, J.

(2025). *Augmented Data Low Confidence (ADLC): A confidence-driven data augmentation framework with ensemble optimization for enhanced machine learning performance*. *IEEE Access*, 13, 201439–201459.

<https://doi.org/10.1109/ACCESS.2025.3636729>

Rane, N. L., Mallick, S. K., Kaya, O., & Rane, J. (2024). Applications of machine learning in

healthcare, finance, agriculture, retail, manufacturing, energy, and transportation: A review. In *Applied Machine Learning and Deep Learning: Architectures and*

Techniques (112–131). Deep Science Publishing. <https://doi.org/10.70593/978-81->

[981271-4-3_6](https://doi.org/10.70593/978-81-981271-4-3_6)

United Nations Framework Convention on Climate Change (UNFCCC). (2025). *Artificial intelligence for climate action: Opportunities for adaptation and mitigation*.

https://unfccc.int/ttclear/misc_/StaticFiles/gnwoerk_static/AI4climateaction/f2922b97c4cf431996c468e622127eb5/112f8be560ea447dab5ff2e53ab3f6e4.pdf

Forbes Advisor India. (2024, February 6). *Top AI statistics and trends: Impact of AI on industries including manufacturing*. *Forbes*.

<https://www.forbes.com/advisor/in/business/ai-statistics/>